

SINR Map for a 5G Urban Macro-Cell Test Environment

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1. Abstract:

In this paper, we simulate the construction of a 5G macro-cell test environment that can visualize the signal-to-interference-plus-noise ratio (SINR) for each antenna overlaid on a real city map. This project adopts ISM and 5G wireless communication. The antennas are designed using antenna toolbox along with phased array system toolbox in MATLAB. The test environment is based on a dense urban area with high user density and traffic loads, focused on pedestrian and vehicular traffic (Dense Urban-eMMB), located in the Romanian city of Focșani. Our antennas are arranged in a hexagonal network cell implemented using Phased Array System Toolbox.

2. State of the art:

2.1. Evolution of Standards and Services

The evolution of 5G New Radio (NR) is mainly driven by 3GPP standards [0]. It began with Release 15 in mid-2018 and has progressed through Releases 16, 17, 18, and 19 to improve network capabilities [1]. Unlike earlier generations that focused on human users, 5G emphasizes applications for people along with the automatic connection of billions of devices through Machine-to-Machine (M2M) and Internet of Things (IoT) services. To support these services, modern networks aim for strict Key Performance Indicators (KPIs), which include 99.999% availability, 99.99% reliability, peak bitrates of around 20 Gb/s, and ultra-low latencies [3, 4].

2.2. Spectrum Utilization and Network Densification

While the first 5G deployments used mid-frequency bands (3–6 GHz) to complement existing 4G infrastructure, current research and deployments are moving toward high-frequency millimeter-wave (mmWave) bands (24–100 GHz). These high bands offer a much wider spectrum of up to 400–800 MHz, which is needed for massive capacity. However, the high loss in signal strength at these frequencies requires a change from large macrocell setups to dense small-cell deployments. In this model, individual cell sizes are limited to about 100 meters, so hundreds of small cells are needed to match the coverage area of a single 4G microcell [2].

2.3. Advanced Antenna Systems and Beamforming

A key part of modern 5G planning is using Massive multiple input multiple output (MIMO) [6, 7] and phased arrays. According to IMT-2020 specifications [5], 5G base stations operating at 30 GHz can have up to 256 antenna elements in one array. These arrays use dynamic beam steering to send radio waves directly to specific users, which helps lower interference from outside the cell. To further improve signal quality, advanced methods like Dolph-Chebyshev amplitude tapering are used to reduce unwanted sidelobe power [9, 10], enhancing the Signal-to-Interference-plus-Noise Ratio (SINR).

2.4. Interference Mitigation and Deployment Planning

To ensure consistent coverage and prevent out-of-cell interference, modern network designs often follow a hexagonal pattern with a minimum Inter-Site Distance (ISD) of 200 meters for cells with a 100-meter radius [8]. Traditional three-sector cell designs are being reconsidered in favor of six-sector setups to provide better RF coverage. Additionally, advanced planning now uses simulation-based methods that include detailed terrain profiles and mathematical models, such as ITU-R P.838-3 for rainfall and P.840-6 for fog/clouds, to account for atmospheric signal loss at high frequencies [11].

3. Theoretical fundamentals:

3.1. Spatial Network Geometry and Sectorization Topology

To model cellular coverage within a realistic urban test environment, base station deployments must be represented through explicit geographic coordinate systems combined with directional sector layouts[14]. In this system, each base station tower i is designated by a spatial vector containing latitude, longitude, and an operational antenna height h_i [15]:

$$p_i = [Lat_i, Lon_i, h_i]^T$$

To expand network capacity and limit co-channel interference through spatial isolation, sectorization is applied independently to each site [19]. Instead of broadcasting uniformly, a tower is partitioned into a discrete number of directional sector transmitters, defined by the per-tower variable $N_{trans,i}$. To maintain spatial balance and provide seamless azimuthal coverage around the site coordinates, the primary steering boresight angles $\phi_{k,i}$ for each sector k are derived by incorporating a half-interval phase offset [15]:

$$\Delta\phi_i = \frac{360^\circ}{N_{trans,i}}$$

$$\phi_{k,i} = \frac{\Delta\phi_i}{2} + k \cdot \Delta\phi_i, \text{ for } k = 0, 1, \dots, N_{trans,i} - 1$$

This half-spacing alignment distributes energy more uniformly across the cell borders. To confine the vertical radiation footprint to ground-level user equipment (UE) and suppress out-of-cell interference bleeding over the physical horizon, each sector incorporates an adjustable mechanical and electrical down-tilt angle, defined per tower as $\theta_{tilt,i}$ [14], [15]

3.2. Microstrip Patch Antenna and Uniform Rectangular Array (URA) Modeling

To achieve the high directivity and fast electronic beamforming required by mid-band 5G configurations, individual sector transmitters discard omnidirectional elements in favor of a Uniform Rectangular Array (URA) constructed from microstrip patch antenna elements [16].

3.2.1. Single Microstrip Patch Element

The base radiating element modeled within the simulator is a microstrip patch antenna physically designed to resonate at the target 5G mid-band carrier frequency $f = 3.0 \cdot 10^9 \text{ Hz}$. The corresponding free-space wavelength λ is governed by the electromagnetic wave propagation constant:

$$\lambda = \frac{c}{f}$$

To ensure maximum radiation intensity is driven outward along the horizontal boresight vector, a 90° coordinate rotation matrix is applied to the element's axis relative to the z -axis, orienting the array normal vector face along the x -axis.

3.2.2. Single Microstrip Patch Element

By arranging the identical patch elements into a grid, constructive and destructive wave interference can be leveraged to shape high gain directional beams. The total radiation of ontenstity pattern $G(\theta, \phi)$ of the URA is determined by the principle of pattern multiplication, expressed by the following formula:

$$G(\theta, \phi) = |AF(\theta, \phi)|^2 \cdot G_{element}(\theta, \phi)$$

Unlike conventional arrays that use uniform excitation currents, this implementation introduces amplitude beam-tapering to suppress side-lobes [9], [10]. The simulator dynamically computes non-uniform amplitude weights using a two-dimensional Dolph-Chebyshev window function matrix. The tapering weights along the vertical array axis (w_m) and horizontal array axis (w_n) are calculated as a function of the targeted side-lobe attenuation parameter, resulting in a composite array factor:

$$AF(\theta, \phi) = \left[\sum_{m=0}^{N_{row}-1} w_m e^{jm(kd_{row} \sin \theta \cos \phi + \beta_x)} \right] \cdot \left[\sum_{n=0}^{N_{row}-1} w_n e^{jm(kd_{row} \sin \theta \cos \phi + \beta_y)} \right]$$

Where $k = \frac{2\pi}{\lambda}$ is the wavenumber, and β_x, β_y are progressive phase shifts used to electronically steer the beam down-tilt. By controlling the amplitude distribution through w_m and w_n side-lobe emissions are minimized, preventing signal leakage into adjacent sectors.

3.3. Radio Wave Propagation and Close-In Path Loss Model

To calculate signal coverage across a city landscape, the system employs an empirical propagation model. As defined within the visualization scripts, a Close-In (CI) reference distance path loss model is used to evaluate signal decay over the physical distance vector [17]:

$$PL_{CI}(d_0) = PL_0(d_0) + 10n \log_{10} \left(\frac{d}{d_0} \right) + X_\sigma$$

where d_0 is the close-in reference distance set to 1 m. The baseline free-space path loss $PL_0(d_0)$ at this reference point is governed by the Friis transmission equation at the 3.0 GHz carrier frequency [17], [18]:

$$PL_0(d_0) = 20 \log_{10} \left(\frac{4\pi \cdot 1 \cdot f}{c} \right) = 20 \log_{10}(40\pi)$$

3.4. SINR Theory

The fundamental metric used to quantify radio link quality, data throughput, and overall coverage viability at any coordinate position on the map is the Signal To Interference plus Noise Ratio [18].

For any point on the map, the received power $P_{RX,m}$ from an active sector transmitter i is calculated using a link budget calculation [15], [18]:

$$P_{RX,i} = P_{TX,i} + G_{TX,m}(\theta, \phi) - PL_{CI}(d_m) + G_{RX} - L$$

Where $P_{TX,m}$ is the per-tower transmission power configuration ranging from 30 to 46 is $G_{TX,m}(\theta, \phi)$ is the directional gain generated by the tapered URA, G_{RX} is the receiver antenna gain, and L represents system feeder losses [15].

3.4.1. SINR Formulation

Assuming a user equipment (UE) terminal dynamically connects to the sector providing the strongest received signal power (the serving cell), all remaining concurrent transmissions on the same 3.0 GHz channel act as co-channel interference. The linear SINR is defined as [15], [18]:

$$SINR = \frac{P_{RX,serving}}{\sum P_{RX,j} + N}$$

$$N = -174 \frac{dBm}{Hz} + 10 \log_{10}(B) + NF$$

Where $\sum P_{RX,j}$ represents the cumulative sum of co-channel interference power received from all other active sectors across the network. The variable N represents the additive background thermal noise power floor over the designated channel bandwidth, modeled using the standard thermal noise formulation and NF is the operational noise figure of the receiver hardware [15], [18].

3.5. Algorithm Multi-Object Optimization Framework

The network design avoids uniform global parameters by using a Genetic Algorithm (GA) metaheuristic to optimize a multi-variable chromosome vector X . For a network containing N_{towers} the chromosome length scales as 5 times the towers. The design vector for each individual tower i is defined as:

$$X_i = [P_{TX,i}, N_{trans,i}, \theta_{tilt,i}, N_{row,i}, dB_{down,i}]$$

The optimization engine navigates this combined continuous and integer-constrained search space to maximize a multi-objective scoring function [19]. To prevent local outliers from dominating the ecosystem, decibel metrics are compressed into a stable range using a hyperbolic tangent (tanh) function.

$$Score = 0.4 \cdot C + 0.25 \cdot \tanh\left(\frac{\mu SINR}{10}\right) + 0.25 \cdot \tanh\left(\frac{SINR_{th}}{10}\right) - 0.1$$

4. Implementation and results:

4.1. Simulation Environment and Software Architecture

The 5G urban macro-cell test environment is implemented in MATLAB utilizing the Antenna Toolbox, Phased Array System Toolbox, and Global Optimization Toolbox. The software architecture centers around an interactive Graphical User Interface (*ControlGui.m*), which decouples user input from the underlying computational optimization back-end. Through the interface, a network designer interactively plots any arbitrary number of base station tower locations onto a high-resolution topographic map centered on the city of Focșani, Romania. The GUI assembles these coordinates along with operational antenna installation heights into a structural tower matrix. This array is passed directly to the simulation engine via *generateSim.m*, which acts as the control center for a metaheuristic search loop designed to automatically discover the optimal configuration for each individual transmitter.

4.2. Algorithmic Optimization

To overcome the limitations of standard grid-search techniques, the simulator implements a Genetic Algorithm (GA) via MATLAB's *ga* optimizer. This approach replaces uniform global parameters with a completely heterogeneous network optimization layout. The solution space is defined by a continuous and integer-constrained chromosome design, where the total number of design variables (*nvars*) scales dynamically. For each tower, the GA simultaneously optimizes five independent variables: transmitter power bounded from 30 to 46 dBm, sectorization order from 3 to 8 sectors, mechanical/electrical down-tilt from 0 to 20 degrees, antenna array row size from 4 to 16 elements, and side-lobe tapering attenuation from 20 to 40 dB.

The mixed-integer profile identifies the index boundaries dynamically within an index array logic. The fitness calculation layer is driven by *evaluateGA.m* and *decodeGA.m*. To handle the massive computational load of simulating multi-generation wave propagation, *computeSINR.m* introduces a highly optimized geographic grid with a resolution step of 0.0004 degrees (~28 meters) over a 1 km x 1 km area. Furthermore, the receiver array coordinates are managed within a persistent memory block. Caching these receiver positions prevents MATLAB from repeatedly instantiating objects, drastically increasing execution speed and protecting system memory across hundreds of GA generations.

4.3. Performance Evaluation and Metric Scoring

Once a network configuration is constructed, the system calls *computeSINR.m* to run the RF propagation layer over the city layout. The raw SINR data points are compiled into four key performance metrics: Global Coverage, Mean SINR, Cell-Edge SINR and SINR Standard Deviation. These metrics are fed into *scoreNetwork.m* which uses a hyperbolic tangent soft normalization function to compress the unbounded decibel metrics into stable bounded values between -1 and 1, ensuring extreme local performance spikes do not skew the global network evaluation. The score is calculated using the following:

$$Score = 0.4 \cdot C + 0.25 \cdot \tanh\left(\frac{\mu SINR}{10}\right) + 0.25 \cdot \tanh\left(\frac{SINR_{5th}}{10}\right)$$

4.4. Simulation Results and Discussion

The Genetic Algorithm successfully navigated the highly complex, multi-dimensional search space, converging on an optimized heterogeneous network profile. The resulting layout demonstrated the profound benefits of per-tower parameter tuning over uniform deployments. In dense central zones vulnerable to co-channel interference, the GA engine automatically converged on highly directive 16x16 arrays paired with aggressive Dolph-Chebyshev tapering and down-tilts up to 20 degrees. This combination successfully narrowed the main radiation lobes, isolating overlapping sectors and significantly raising the cell-edge SINR percentile. For peripheral or widely separated tower nodes, the algorithm adjusted by selecting higher transmission powers up to 46 dBm and lower down-tilts to maximize geographic coverage. By matching the antenna array scale and power output to the specific spatial surroundings of each base station, the GA framework effectively balanced core capacity and boundary link stability across the entire urban footprint of any city.

5. Conclusion:

This project successfully developed an interactive 5G network optimization simulator in MATLAB to model and analyze a 3.0 GHz urban macro-cell deployment. By integrating an intuitive user interface with advanced toolboxes, the system allows network planners to seamlessly input spatial site coordinates and automatically evaluate network layouts. The core contribution of this work is its automated Genetic Algorithm (GA) metaheuristic optimization engine, which replaces subjective tuning and rigid global configurations with a fully heterogeneous, per-tower variable design. By applying a hyperbolic tangent soft-normalization technique to raw SINR data within an optimized multi-objective scoring framework, the algorithm effectively balances the competing demands of maximizing global coverage and core spectral efficiency while minimizing cell-edge dropping risks and signal variance.

Future extensions of this work could enhance the simulator's realism by replacing the empirical Close-In path loss model with dynamic 3D ray-tracing engines that account for explicit building heights and material reflections. Additionally, the current centralized Genetic Algorithm framework could be upgraded to advanced multi-objective evolutionary algorithms, such as the Non-dominated Sorting Genetic Algorithm II (NSGA-II), to map out true Pareto-optimal fronts between deployment overhead and network performance. Finally, incorporating dynamic user traffic profiles and moving into mmWave frequencies would allow the simulator to model the dense, fluctuating user demands characteristic of next-generation heterogeneous networks.

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